

Embedded Firmware

Engineering Technology

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Embedded Firmware (A04604)

Short Title:	Embedded Firmware	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Electronics	
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Credits	5	Level: Advanced

Description of Module / Aims

This module builds on concepts introduced in Embedded Systems Architecture & Hardware module. It covers the programming of high performance embedded processor systems using C and assembly language. Software development tools are extensively used with the main emphasis being on application development, testing, debugging and verification.

Programmes

COME-0004	BEng (Hons) in Electronic Engineering (WD_EONIC_B)	4 / 8 / M
COMP-0970	BEng (Hons) in Information Engineering (International) (WD_EEELC_BI)	4 / 8 / M
COMP-0970	BSc (Hons) in Applied Computing (WD_KACCM_B)	4 / 8 / E
COMP-0970	BSc (Hons) in Applied Computing (WD_KCOMP_B)	4 / 8 / E
COMP-0970	BSc (Hons) in Computer Science (WD_KCMSC_B)	4 / 8 / E
COMP-0970	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	4 / 8 / E
COMP-0970	BSc (Hons) in the Internet of Things (International) (WD_KINTT_BI)	4 / 8 / M

Indicative Content

- Architecture of Embedded Systems and programming model.
- Instruction set and assembly language programming.
- Subroutines, parameter passing, and good programming practice
- Exceptions and interrupts, Vectored Interrupt controller (VIC), C and assembly language constructs for Structured Programming.
- APCS compliant modules , Mixed programming/translation in C and assembly
- Polling and interrupt driven I/O, accessing and programming I/O devices, timer devices.
- Software development tools, Application Programming Interface (API), libraries.
- Program design, modular programming, testing, debugging and verification, documentation and maintenance

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Demonstrate knowledge and understanding of processor's instructions and its peripheral devices.
2. Create programs in Assembly language, C language and mixed languages using APCS standard.
3. Analyse and translate C language software into Assembly language and vice-versa.
4. Design application programs, and document it, in C/Assembly language using structured and modularised approaches.
5. Use software development tools to write, compile, test and debug programs.
6. Create and use software libraries and API.

Learning and Teaching Methods

- Lectures and tutorials
- Practicals
- Mini-Project (Teamwork)

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1, 2, 3
Final Practical	40%	4, 5, 6, 7

Assessment Criteria

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Furber, S. B. *_ARM System On Chip Architecture_*. 2nd.. UK: Adison-Wesley, 2001 (Seminal).

Supplementary Material

- Smith, W.A. *_C Programming for Embedded Microcontrollers_*. 2nd.. UK: Elektor International Media BV, 2008.

Requested Resources

- ENGINEERING LAB: Electronics